

# READING ACCESSIBLE NEWSPAPERS



Swallowed by  
sand A house in  
the fishing village  
of Sheyna,  
Russia...

Memories of a war  
that wasn't a war

Missile sites  
suggest deceit by  
North Korea

Saudis talked of  
assassinations

Dating game with  
billionaires as  
contestants



— LOD +

1

## LOW VISION: CONTEXT AND NEEDS



*Age-related Macular Degeneration is the primary cause of visual deficiency in industrialised countries*

## Reading with a central vision loss

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Low vision is a societal challenge. Global estimations show a gradual increase in the absolute number of people who are blind or have moderate to severe visual impairment (MSVI) from 253 million people in 2015 to 703 million people in 2050 (projection, from IAPB Vision Atlas).

### Age-related Macular Degeneration

Low vision refers to a significant visual impairment that cannot be corrected fully with glasses, contact lenses, medication or eye surgery. Retinal diseases are a common cause of low vision, resulting in different kinds of impairments. Among retinal diseases, Age-related Macular Degeneration (AMD, "DMLA" in French) is the primary cause of visual deficiency in industrialised countries. It is considered an incurable disease at this point. It causes the destruction of the macula which is the area surrounding the fovea, near the centre of the retina in the eye. As these patients are blind in their central visual field (region of keenest vision), many perceptual functions are dramatically degraded which induces an important loss of autonomy, a degraded quality of life, and often a severe depression.

### Macular degeneration is widespread

Age is a prominent risk factor for AMD. The risk of getting advanced AMD increases from 2% for those ages 50-59, to nearly 30% for those over the age of 75. In France, INSEE

estimates that there are 3.1 million people with visual impairment, and already 1.5 millions with AMD. Because of the ageing of the population, the number of people affected by AMD is expected to increase significantly in the years ahead.

### Causes are not well understood

Macular degeneration may be hereditary, meaning it can be passed on from parents to children. Other factors related to your lifestyle such as smoking, diet, or high blood pressure, may increase the likelihood of macular degeneration. However, the precise causes are not well understood and so far, while several medical treatments are available (for wet macular degeneration), none can cure the condition.

### Reading is a "must have"

Reading is amongst the strongest need reported by patients. Problems are distortion of letters and lines, loss of visual acuity, and central blind zone (scotoma) masking small letters and words. As a consequence, while people leaving with AMD can still perform a variety of other daily leaving activities, losing this ability to read when the vision pathology evolves brings them frustration, anxiety, and nervousity, and can lead to both strong psychological and cognitive consequences.

### Rehabilitation

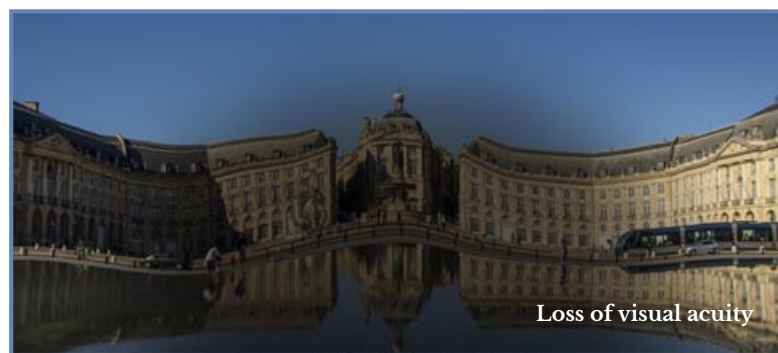
Although AMD is an irreversible macular retinopathy, patients still have a plastic and functional vision in their peripheral visual field.

Patients can develop efficient "Eccentric Viewing" to adapt to the central blind zone (scotoma), i.e., they can learn to direct gaze away from the object or text they want to identify thanks to occupational therapists and other rehabilitation professionals. The development and retention of a so-called preferred retinal locus (PRL) for fixation provides improvements in the reading performance of patients in a majority of cases. However, these improvements are generally insufficient for a good reading comfort due to limited reading speed and endurance.

### Accessible reading devices and media

To further help patients, palliative solutions are available in the form of reading aids. The majority of current solutions focus on helping people to read printed material through magnification. For reading printed material, electronic magnifiers have offered new possibilities for the patients to tune their preferred display and many studies have been done to understand the impact of most parameters in reading performance. For reading electronic documents on digital devices such as e-readers or iPads they provide low-vision users with interesting features such as intuitive touch screen gestures (e.g., pinch-to-zoom) and vision aids that support inverted colours. For online content, a number of screen magnification programs (e.g., ZoomText, SuperNova) are also available to users.

Worldwide, the number of people living with AMD is expected to reach 196 million worldwide by 2020 and increase to 288 million by 2040.



Loss of visual acuity



Apparition of central scotoma

Symptoms of the apparition and evolution of AMD, simulating the perception of a patient

(Source: <http://tiny.cc/auia9y>)

## Reading newspapers

Independently of the media used to access content, magnification is the main solution. Magnification improves reading ease for low vision patients mostly for linear text passages such as books or single articles. However, finding efficient solutions to read complex documents such as newspapers remains an open challenge. One problem comes from the limited screen space to show magnified content, so that one faces the difficulty of both global (e.g., between articles, sections, media content) and local (e.g., line to line) navigation. This will be the core motivation of our project that will be described in the following pages. But before entering the subject matter, let us remind why reading newspapers is so important, and what is available so far to make news accessible to low vision people.

### Become part of a well evolved society

Of course, newspapers provide news about a country's economic situation, sports, games, entertainment, and commerce. However, many readers feel the newspaper is important not just because it gives them information or is a relaxing activity, but also because it enables them to appear more informed at social gatherings, exchange with others on current stories and in the end feel reassured of their position and existence in their environment. In other words, newspaper help you become part of a well evolved society. These social reasons explain the psychological impact encountered by disabled people that are progressively loosing this link to the society. This is why ensuring equal access to content is essential.

### A unique accessibility challenge

Unfortunately, newspapers pose a significant accessibility challenge due to a number of reasons, particularly (1) complex design layout, making it hard to navigate with traditional hand-held magnifiers (local/global navigation problem), (2) temporal nature, which does not allow extra time for publishers to produce accessible versions, and (3) non-standardised formatting between different publishers, which poses challenges to develop automated approaches that adapt the content with more accessible design.

### A design challenge...

The digital era is completely transforming the newspaper industry, offering new user experience for all readers but also new opportunities for low vision people to access information. However, to be successful, newspaper designers now stand before a great design challenge: translating the design from the printed edition and the functionality from the online edition into this new technology, making a synthesis based on usability. This is what our project is about. ■

## From print to web: How low vision people access news?

### Newsprint



### Large print edition



### Web edition, online



**Newsprint** is the traditional way to read news. Because of the complex layout and small fonts, magnification equipments are in general necessary.

Because magnification may not be sufficient, there is an offer of **large print editions** which are mainly limited to a selection of books. For the

press, the only existing case is the New-York Times. They propose a weekly 40-page publication that is published in 16-type font, and is a summary of the previous week's Times. Finally, one can read **web editions, online**. The W3C Web Accessibility Initiative as provided

principles for content providers, designers, and developers. However, because these recommendations are not always properly implemented, screen readers (e.g., Jaws, VoiceOver) may encounter some difficulties to deal with advertisements and cluttered website design.

*Because newspaper design is about user experience and engagement of the readers, our goal is to develop a new digital offer that could also fit vision impaired people needs.*

# Making Newspapers Accessible again through accessible design

The digital era has completely transformed the newspaper industry which is still struggling for finding the best revenue models, stretched between printed and online news. As print ad revenue cratered, the need to squeeze revenue from digital sources grew and pushed publishers to invest on developing online news content, again making up the money in advertising.

## Online news are not the solution

Online news offer a number of advantages and interesting characteristics to the users such as, e.g., continuous updates, the possibility of searching, intuitive navigation, hyperlinks, and of course the possibility to read news mostly everywhere with any portable device. However, the quality gap between online news and printed newspaper remains very large. This is due to three main reasons. The first reason is technical and related to the fact that website should be legible on any platform. This is made thanks to responsive web design principles, allowing the

layout to respond to the user behaviour and environment based on platform, screen size, and orientation. However, making sites that can refactor themselves for small screens implies a strong simplifications in the structure of informations. Websites mostly present news in a linear fashion, thus missing the editorial organisation found in printed newspapers.

The second is directly related to the ad revenue model that pushed news websites to their breaking point. There is a situation now where ads, even on reputable sites, occupy an important portion of the pages. Some professional in the field even wonder if the journalists still matter about how to deliver news to human beings or instead how many advertisements are served.

The third is related to the accessibility of the content. The W3C Web Accessibility Initiative (WAI) has provided principles for content providers, designers, and developers. However, because these recommendations are not always properly implemented, screen readers may encounter some difficulties to deal with cluttered

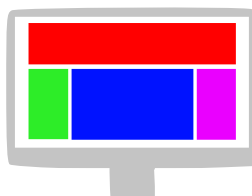
website design and advertisements, which are very common in news websites.

## Back to print newspapers design.

This naturally brings us back to print newspapers for which user experience is a central point. Indeed, despite its decline, print newspapers continue to be an indispensable point of reference in terms of quality. By quality, here we refer to newspaper design, that is having a strong, consistent and appealing visual approach to enhance the impact and comprehension of a story.

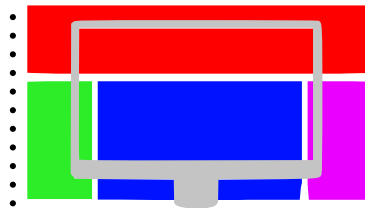
Our project aims at developing computational models for newspaper design. By definition, newspaper design consists in having a strong, consistent and appealing visual approach to enhance the impact and comprehension of a story. The quest of beautiful design is not a cult of "art for art's sake" advocated by a few extreme visual experts. Neither is it just about making things look good using

## What this project is NOT about

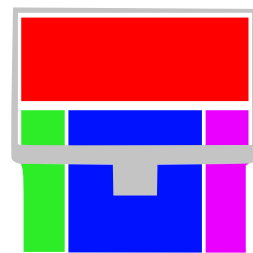


### Responsive design

The goal of responsive design is to build web pages that detect the visitor's screen size and orientation and change the layout accordingly. It is defined and implemented by web developers through, e.g., style sheets. Nothing is automatic. Purpose is not accessibility.



### Magnification



Magnification (of font) has been identified as the main solution to facilitate reading for low vision people. However, due to limited screen space, magnification produces an overflow of the content, resulting a local/global navigation problem. Design can also be ruined by this operation.

## Proposed breakthrough: Design preserving magnification



### Illustration of our approach

On the **left**, we show how an original newsprint front cover fit into an e-Reader: it is hardly readable due to printsize.

On the **right**, we show how we plan to present the same page (here, manually built layout). Given a magnification factor our approach aims at producing automatically a complete redesign of the page according to newspaper design rules, which also fit user needs in terms of visibility. This redesign can be seen as a packing problem looking for the best positions and shapes of the blocks of text. A hierarchical description of the page will also be used to decide which elements to preserve or cut, following priority rules. Note that, for images, image manipulations (e.g., content-aware cropping) will also be part of the final solution we envision.

proper grids, type, colour, and visuals. Newspaper design is about user experience and engagement of the readers. In other words, by focusing on design, we address the overall reading experience.

### What about magnification then?

When reading newspapers using magnification, users face the local/global navigation problem. It is a general navigation problem across scales which is even more detrimental for visually impaired. As readers want to read, they need to magnify to access large scale, i.e., a smaller amount of area with a greater amount of detail (larger print size). When going back to the small scale (showing the full page), the reader may encounter difficulties to know what was read at large scale and what remains to be read. The main reason for this problem is the fact that magnification results in focusing on a smaller amount of area from the small scale.

### Proposed breakthrough

Our project intends to bring another solution to solve the local/global navigation problem due to magnification which consists in keeping the content from small scale and the details of the large scale. The difficulty is that one faces conflicting constraints, namely retaining the editorial policy (i.e., newspaper page content) accommodating the device's specifications and matching the reader's visual capabilities (i.e., visual acuity). To meet these constraints, we foresee an original approach using combinatorial/geometric optimisation. Our ambition is to design a method that automatically generates newspaper layouts depending on a magnification factor, with a high visual quality, keeping storytelling unaltered and preserving the reader's experience. We refer to this objective as design-preserving magnification.

#### A three-step approach

We foresee a three step approach to automatically generating newspaper layouts (see grey box).

#### A three-step approach

- Step 1: Page understanding to obtain a hierarchical description of the page, via segmentation into blocks, classification and extraction of design rules.
- Step 2: Page re-laying-out, via block packing under priority rules with constraints related to newspaper design, aesthetics and of course user preferences in terms of font size readability.
- Step 3: Page filling-in, compliant with media size and visual acuity constraints.

#### A new packing problem

The bottleneck is Step 2, as it hinges on a difficult combinatorial/geometric packing problem. Step 2 is the focus of the proposal, and solving it will pave the way to a complete solution to the redesign problem. This Step 2 is a new design-to-design problem, that is producing aesthetically pleasing layouts from another existing layout requiring a reorganisation of the structure, raising a packing problem. Given each block of text

and images, a finite set of admissible shapes can be inferred, given constraints related to aesthetics and print size; then one needs to pack these shapes – a Tetris like problem where one shape per block must be selected.

### NP-complete problem

Our formulation of the problem as a packing problem leads to an NP-hard optimisation problem of a new kind, requiring non trivial algorithmic solutions and associated complexity studies.

### Very high number of variables...

We were able to coin three different Integer Linear Programs (ILP) to solve the problem. These ILPs differ by the encoding of the variables and associated constraints ensuring the layout is valid and of good quality. Interestingly, our ILPs exhibit very different number of variables and constraints while solving the same problem. However, **the number of variables is at least  $10^5$  so that the complexity of solving this ILP is of order  $10^{15}$  which is not tractable.**

### Computing power constraint

The expected result of this project is

a first proof-of-concept of digital newspaper with design-preserving magnification. As such, we will need to consider the efficacy of our algorithm which should run on classical android tablets.

### Visual acuity constraints

Last but not least, another crucial constraint will be visual acuity of the patients and their ability to read depending on the font size. The evaluation of reading performance as a function of font size, although it is based on simplified reading scenarios, could provide a first indications to determine the degree of magnification of the fonts which will have a direct impact on the final design. By running fast, users will have the possibility to adjust the presentation according to their preference.

### Funding request and workplan

For this project, we have identified algorithmic and engineering work. The funding request is 125 keuros. for the following workplan:



1 Postdoc,  
18 months:  
75K EUR

+



1 Engineer,  
12 months:  
50K EUR

=



Total budget:  
125K EUR

#### Postdoc (M1-M18)

- Determining the complexity class (FPTAS, PTAS, APX, APX-hard) of the optimisation problem (6 months).
- Code prototyping, and testing relaxed versions of the ILPs and exploring branch-and-bound algorithms, investigating the connexion with SAT solvers (6 months).
- Testing on representative journal types (Le Monde / NYT, L'Equipe, Le Canard) with normal sighted and low vision people (6 months).

#### Engineer (M7-M18)

- Making operational the full pipeline, from a given design to the re-design, integrating and optimising the packing algorithms (9 months).
- Building experimental protocols on representative journal types (3 months).

#### Funding request

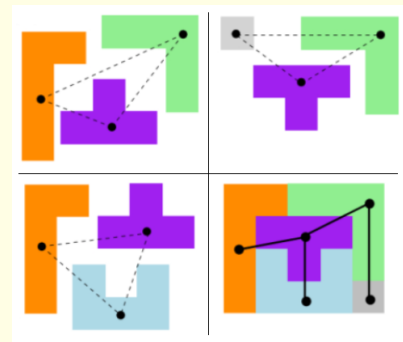
## About us

One major originality of this project is to combine methods from vision used within the Inria BIOVISION team and methods from graph theory and optimisation developed in the Inria ABS team in the domain of computational structural biology. Interestingly, a surprising connection is the following one: let us consider a block as a protein, and a shape for this block as a conformation of its protein. The set of blocks is tantamount to a protein

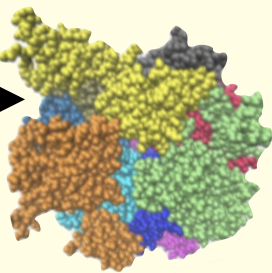
assembly. The problem can be viewed as determining assembly structures (i.e., a conformation for each protein) in order to optimise some global function... Given this combination of skills, our approach is likely to be unique.

We have also two partners from the neuroscience and clinical side (LPC and CHU Pasteur) who will contribute to the evaluation of our approach with low vision patients.

Contact: pierre.kornprobst@inria.fr,



Connectivity inference between proteins of a large molecular assembly (Source: EPI ABS, SBL, <http://tiny.cc/rooi9y>)





### Consortium



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*Digital innovation could be the answer to save the press industry from its decline*

## Design preserving magnification:

# A digital innovation to serve the press industry?

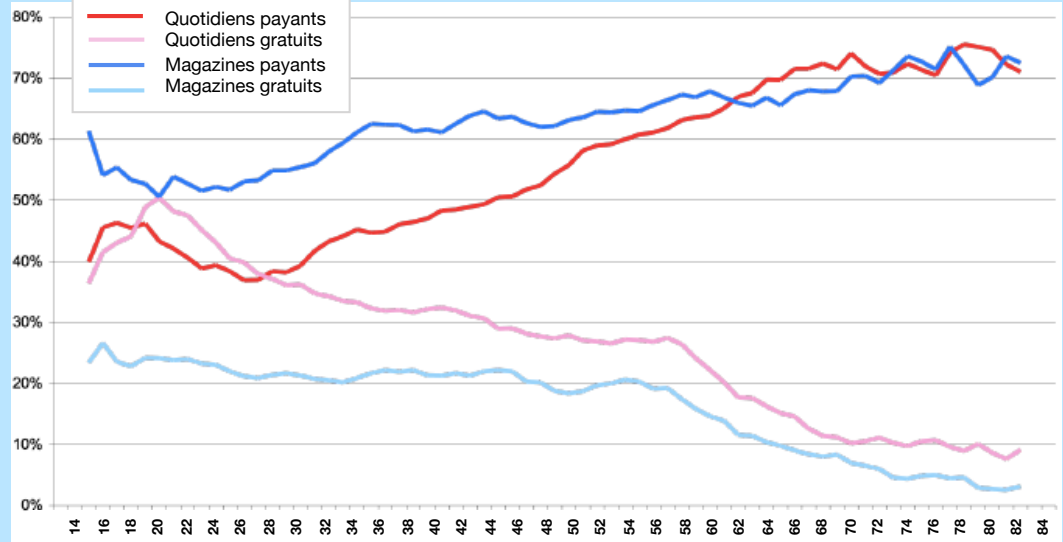
Faced with a structural crisis since the 2000s, the press faces major technological and economic challenges: the development of information media and online services, competition from free titles have permanently weakened the sector. With a turnover of 7.5 billion euros in 2017, newspaper publishers have lost 42% of their revenues since 1990. This decrease is explained by the decline in sales of paper copies, combined with that of advertising revenue. Over the period 1990-2015, advertising investments decreased overall by 61% (-59% for commercial advertising and -68% for classified ads). However, despite the decline in single-copy sales, subscription revenue grew by 27% between 1990 and 2017.

## Seniors habits help the press industry to face the decline

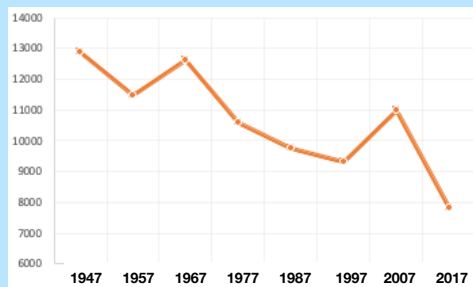
When the press industry will start to look for news subscribers it will be important to highlight that most people paying currently for subscriptions are people between 50 y.o and 80 y.o, those people with a strong printed content culture and attached to traditional media are an interesting target for newspapers. 50% of 65+y.o people buy a newspaper everyday and spend twice as much reading time as the rest of the French population.

## The digital potential

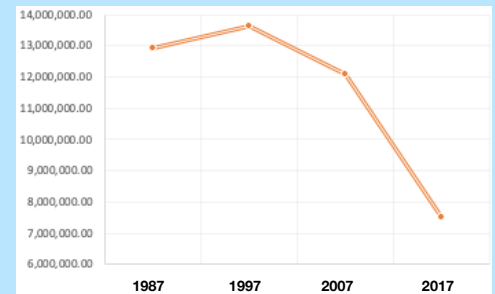
For many years, newspaper publishers have simply wanted to exist on the web. This late awareness of the digital potential is slowly starting to pay, digital sales are progressing and these results encourage all industry players. In Europe, the digital shift in daily newspapers has become a reality with e-newspapers. For example, The Guardian had seen its circulation shrink considerably from 358,844 copies in 2009 to 207,958 copies in 2014 and 152,714 in 2018. With the launch of a new site in 2015, it remains one of the most read with more than 150 million



Evolution of reading habits per age (Source: APPM/ Etats Généraux de la Presse)



Total printed copies in the french market (Source: Dir. générale des médias et des industries culturelles)



Revenues of the french press industry (Source: Conseil supérieur des messageries de presse.)

unique visitors per month, thanks to a globalised two-thirds readership. In 2018, for the first time in its history, digital revenues surpassed those from paper: £108.6m for £107.5m. In any case, and a little bit late, the revolution is just beginning for the press industry. The digital world will continue radically transforming the entire value chain of this industry, but above all, it will transform how people are informed and how the media communicate.

**50.3%**  
The digital revenues from The Guardian

## Perspectives of our project

Beyond our original motivation to make newspapers accessible again for low visually impaired people, this project has a potential to lead

to two types of innovative solutions for:

- **Re-layouting:** Our solution could enrich the reference editorial (re-)layout software, to be used by all major stakeholders (e.g., as a plugin to reference DTP software).
- **Layouting from scratch/ personalised:** As a further step, we shall investigate our tools and methods to design page layout presenting material related to a specific topic (or a small set thereof). This goal is of particular interest for personalised layout, thanks to Artificial Intelligence in order to learn features most suited for a specific class of readers. ■

## READING ACCESSIBLE NEWSPAPERS



### THE MOTIVATION

An estimated 217 million people live with moderate to severe vision impairment. Age-related macular degeneration (AMD) alone represents 45 percent of all cases of low vision, with one out three people after the age of 75 concerned by this pathology. Among all consequences, because AMD affects central vision, reading abilities become severely reduced. Losing the ability to read brings people frustration, anxiety, nervousity and can lead to strong psychological but also cognitive consequences. With no cure available so far, one should invent novel palliative solutions to help them read again with more ease and pleasure.



### THE NEED

Reading newspapers is considered as an important activity not only to be informed or because it is a relaxing activity, but also to feel part of a well evolved society. This explains the psychological impact encountered by disabled people that are progressively losing this link to the society and this is why ensuring equal access to content is essential. Unfortunately, newspapers pose a significant accessibility challenge due to a number of reasons notably related to the complexity of the design. Finding efficient solutions to read complex documents such as newspapers remains an open challenge.



### THE SOLUTION

Magnification (of font) has been identified as the main solution to facilitate reading for low vision people. However, due to limited screen space, magnification produces an overflow of the content, resulting in a local/global navigation problem. Our project intends to bring another solution to solve the local/global navigation problem due to magnification which consists in keeping the content from small scale and the details of the large scale. The difficulty is that one faces conflicting constraints, namely retaining the editorial policy (i.e., newspaper page content) accommodating the device's specifications and matching the reader's visual capabilities (i.e., visual acuity). To meet these constraints, we foresee an original approach using combinatorial/geometric optimisation. Our ambition is to design a method that automatically generates newspaper layouts depending on a magnification factor, with a high visual quality, keeping storytelling unaltered and preserving the reader's experience. We refer to this objective as design-preserving magnification.



### THE EVIDENCE

One major originality of this project is to combine methods from vision used within the Inria BIOVISION team and methods from graph theory and optimisation developed in the Inria ABS team in the domain of computational structural biology. Given this combination of skills, our approach is likely to be unique.



### THE CALL-TO-ACTION

The expected result of this project is a first proof-of-concept of e-newspaper with design-preserving magnification. To implement this project, we have identified algorithmic and engineering work. The cost estimate is 125 keuros, for a 18-months project, covering the recruitments of a postdoc and an engineer.



### THE PERSPECTIVE

Beyond our original motivation to make newspapers accessible again for low visually impaired people, this project has a potential to serve the press industry in offering high quality content across devices.



### THE CONTACT

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